transition risk that will materially impact its business, results of operations, or financial condition in the short- or long-term and incorporate risks associated with such future cash flows into asset values today. Indeed, academic literature shows that risks both in the near term and far into the future are priced into current asset valuations.[2824] Thus, for many registrants, GHG emissions can be helpful to assess the registrants' exposure to climate-related risks, particularly to material transition risks.[2825]

As noted in section IV.A, many registrants currently do not provide quantitative disclosures on their Scope 1 and 2 emissions. This lack of information on emissions makes it more difficult for investors to assess the degree of risk in individual companies, to compare those risks across companies, and to value securities. By requiring disclosure of GHG emissions for specified registrants for the same historical periods as those included in the financial statements in the relevant filing, the final rules will help investors develop a more accurate assessment of those registrants' exposure and approach to climate-related risks over time. For example, Scope 1 and Scope 2 emissions disclosure may be relevant to investors' assessment of a registrant's progress made on targets or goals or towards its transition plan.[2826]

[2824] For evidence that points to the pricing of short-term climate-related risks, *see* R. Faccini, R. Matin & G. Skiadopoulos, *Dissecting Climate Risks: Are They Reflected in Stock Prices?* 155 J. of Banking & Fin., Article 106948 (Oct. 2023); Huynh & Xia (2021). For evidence that points to the pricing of long-term climate-related risks, *see* M. Painter, *An Inconvenient Cost: The Effects of Climate Change on Municipal Bonds*, 135 J. of Fin. Econ. 468 (2020); D.D. Nguyen, S. Ongena, S. Qi & V. Sila, *Climate Change Risk and the Cost of Mortgage Credit*, 26 Rev. of Fin. 1509 (2022); Huynh & Xia (2021).

[2825] *See* letters from CALSTRS; Vanguard; Fidelity; BlackRock; CALPERS; and State of NY Office of the Comptroller.

[2826] Research has shown that issuers tend to "cherry-pick" the baseline year (i.e., pick the year with highest emissions within the past few years) when forming an emissions target so that any progress appears in the most favorable light. *See* P. Bolton & M. Kacperczyk, *Firm Commitments*, National Bureau of Economic Research, No. w31244 (May 2023). The final rules will thus benefit investors by helping them identify when such cherry-picking occurs so as to arrive at a more informed assessment about the registrant's progress towards meeting its targets or goals.

The final rules will provide informational benefits beyond those associated with the voluntary disclosure of emissions that may be found in sustainability reports or other places, such as company websites. In particular, the overall mix of information disclosed to the market can be distorted when only a certain subset of companies (e.g., those with lower emissions or those that face lower costs of emissions measurement) have stronger incentives to make voluntary disclosures. The final rules may offset this distortion because disclosure is only required if a registrant determines that its Scope 1 and Scope 2 emissions are material. The materiality qualifier will allow registrants that determine that their emissions are immaterial to avoid the full costs of emissions measurement and disclosure. It will also mitigate the risk that investors could be burdened with large amounts of information that is less relevant for their investment and voting decisions. In addition, mandatory disclosure of Scope 1 and Scope 2 emissions data in Commission filings may deter potential greenwashing that could occur with voluntary disclosures.[2827]

Some commenters questioned the value of GHG disclosures in light of existing requirements for some registrants to report emissions pursuant to the GHGRP.[2828] As previously discussed,[2829] the data available from the GHGRP is generally not suited to help investors understand how a registrant's exposure and approach to managing climate-related risks may impact its future cash flows and profitability for several reasons. First, the GHGRP requires that

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[2827] R. Yang, *What Do We Learn from Ratings About Corporate Social Responsibility? New Evidence of Uninformative Ratings*, 52 J. of Fin. Intermediation, Article 100994 (Oct. 2022); Soh Young In & Kim Schumacher, *Carbonwashing: A New Type of Carbon Data-related ESG Greenwashing* (2021), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3901278 (retrieved from SSRN Elsevier database); V. Kalesnik, M. Wilkens & J. Zink, *Do Corporate Carbon Emissions Data Enable Investors to Mitigate Climate Change?*, 48 J. of Portfolio Mgmt. 119 (2022) ("Kalesnik *et al*.").

[2828] *See supra* section IV.A.3 and letters from Chamber; Elaine Henry; BOK Financial; David R. Burton; Permian Basin; and Petroleum Association.

[2829] *See supra* section IV.A.3.

emissions are reported at the facility-level rather than the registrant-level. Second, suppliers of certain products must report their "supplied emissions," conditional on these emissions exceeding a specified threshold.[2830] Third, GHGRP reporting is limited to U.S. facilities. Some commenters asserted that the GHGRP could not be substituted for the proposed rules given the different disclosure requirements and the different objectives of the two reporting regimes.[2831]

While there are differences between the EPA's GHGRP and the Scope 1 and 2 emissions disclosures in the final rules, we expect that registrants subject to both reporting regimes would face reduced costs of compliance with the final rules to the extent there is overlap between the reporting requirements of the GHGRP and the final rules. As discussed in section IV.A, the GHGRP covers 85 to 90 percent of all GHG emissions in the U.S. and includes those emissions referenced by the GHG Protocol and included in the final rules' definition of "greenhouse gasses."[2832] As such, we expect that entities subject to the GHGRP disclosure and reporting requirements may consequently have lower incremental information gathering costs under the final rules for those emissions already required to be calculated and reported by a registrant pursuant to the GHGRP. For example, because both the GHGRP and the final rules require companies to collect information to report and disclose their Scope 1 emissions, to the extent that the information and reporting activities overlap, registrants subject to both the final rules and the GHGRP may face lower incremental information gathering costs. However, as one commenter

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2830 In addition, as previously discussed, the EPA emissions data only reflects a portion of emissions. *See supra* section IV.A.3. The EPA's emissions data therefore presents challenges for investors to use, especially as the data are made public by facility and not by company. While each facility is matched to its parent company, this company may not be the entity registered with the SEC and thus the reported information may be less relevant to investors. *See also* letter from EPA (containing a tabular comparison of the EPA disclosures to the proposed disclosures).

2831 *See, e.g.,* letters from EPA; and Marathon Oil.

2832 *See supra* section IV.A.3.

noted, "[t]he Commission-proposed regulation is not completely in alignment with the US EPA regulation. Thus, an assessment, plan of action, and implementation of changes will be needed for many companies to be compliant with the requirements of both agencies."[2833] In addition as noted above, this lower incremental cost would only apply to direct emissions from U.S.-based facilities, not registrants' international facilities or operations.

Limiting the disclosure requirement to larger companies (*i.e.,* those with greater resources that tend to be already calculating emissions, as noted in section IV.A) will help to balance the concerns of commenters who stated that the evolving nature of current emissions measurement technologies could impose significant compliance costs on registrants, especially those not currently familiar with reporting this information.[2834]

Although the final rules limit disclosures to circumstances in which emissions are material for registrants, we expect most, if not all, LAFs and AFs that are not EGCs or SRCs will need to assess or estimate their Scope 1 and 2 emissions to reach a materiality determination. As a result, we expect these registrants will, to some extent, need to adopt controls and procedures to assess the materiality of their Scope 1 and 2 emissions and determine whether disclosure is required if they do not already have them in place. Registrants that determine that their Scope 1 and 2 emissions are material may likewise need to adopt further controls and procedures, including measurement technologies and other tools to track and report the information to the extent they do not already do so. The final rules may also affect registrants that currently track and/or report this information.[2835] For example, some registrants may only be measuring some

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[2833] Letter from Marathon Oil.

[2834] *See* letters from Blackrock; Business Roundtable; and Chevron. *See also* Kalesnik, *et al*., *supra* note 2827.

[2835] As we discuss below, the costs for existing registrants who track and disclose emissions will be limited because the final rules enable registrants to continue to use the operational and organizational boundaries they already use to track emissions.

Scope 1 or Scope 2 emissions.[2836] Any investments in systems or technologies to better measure Scope 1 and Scope 2 emissions will improve the quality of available data[2837] on emissions but will also contribute to the direct costs of compliance.[2838]

The benefits of this component of the final rules depend on the extent to which Scope 1 and 2 emissions disclosures are accurate and thus provide reliable reflections of registrants' exposure to material climate-related risks, their management of that risk, and their progress on transition plans and/or targets and goals (to the extent they have them). Several commenters noted that many registrants have had more experience measuring and disclosing Scope 1 and 2 emissions than Scope 3 emissions, and that those methodologies, from their experiences, are well-established and are considered fairly robust.[2839] Nevertheless, according to studies as well as commenter feedback, there may be issues with errors and inconsistencies in voluntary Scope 1 and 2 emissions disclosures.[2840] The final rules will benefit investors by improving the accuracy and reliability of this information—first through requiring registrants to subject GHG emissions disclosures, to the extent they are required to make them, to disclosure controls and procedures;

2836 *See* Kalesnik, *et al*., *supra* note 2827 (noting that many registrants do not fully measure their Scope 1 emissions).

2837 A number of studies have raised concerns about the quality of existing emissions data. For example, one study found that third-party estimates of emissions, which represent a significant fraction of the emissions data available in several existing databases, are materially less accurate than self-reported emissions data by issuers. *See* Kalesnik *et al*., *supra* note 2827. Another study examined emissions data reported to CDP between 2010 and 2019 and found that 38.9% of the reports exhibited disparities between the reported total emissions and sum of reported emissions by various sub-categories. *See* S. Garcia-Vega, A.G. Hoepner, J. Rogelj & F. Schiemann, *Abominable Greenhouse Gas Bookkeeping Casts Serious Doubts On Climate Intentions of Oil and Gas Companies* (working paper, Mar. 2023), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4451926 (retrieved from SSRN Elsevier database).

2838 *See* sections IV.C.1.b and IV.C.3 for additional information on the associated compliance costs.

2839 *See* letters from National Retail Federation; AHLA; and Aerospace Industries Association.

2840 *See* Kalesnik *et al*., *supra* note 2827; Garcia-Vega *et al*., *supra* note 2837; *see also* letter from Calvert ("Research demonstrates about 30% of companies that disclose such information in their own reporting make errors on a regular to periodic basis, despite the well-established rules and systems that already exist to ensure proper reporting of such emissions. In many cases, this appears to stem from a lack of effective internal controls or well-functioning monitoring systems.").

and second, by requiring assurance. The final rules also permit the disclosure of reasonable estimates for Scope 1 and 2 emissions provided that such estimates are accompanied by disclosure of underlying assumptions and reasons for using estimates, which will help investors better understand the metrics that registrants are disclosing.

Scope 1 and 2 emissions may not fully reflect a registrant's exposure to transition risks because some of those risks would only be captured through other metrics such as Scope 3 emissions.[2841] For example, registrants facing similar exposure to emissions-related climate risks may report different Scope 2 emissions levels depending on, for example, whether they pay directly for their utilities (counted as Scope 2) or their leases provide for utilities expenses (counted as Scope 3), or, as another example, whether they have employees who work from home and therefore who do not contribute directly to utilities expenses. Recognizing these limitations, the final rules also require disclosures on methodology, significant inputs, significant assumptions, organizational boundaries, operational boundaries, and reporting standard used with respect to Scope 1 and 2 emissions. These disclosures will provide additional context to help investors understand the disclosures and will enable investors to draw more reliable comparisons across registrants. For example, disclosure of operational boundaries will help distinguish registrants that rely on utilities provided by third parties from those that pay directly for their utilities, which will assist investors in accounting for this difference when comparing reported emissions and thus climate-related risk across registrants.

In a change from the proposal, we are exempting SRCs and EGCs from the GHG emissions disclosure requirements in order to limit the costs of this disclosure requirement for such registrants. This exemption should also mitigate the risk of deterring prospective EGCs or

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[2841] *See* letters from Wellington; and Calvert.

SRCs from conducting IPOs or inducing EGCs or SRCs to deregister under the Exchange Act as a result of the costs associated with compliance with the requirements to disclose material Scopes 1 and 2 emissions. Registrants that already measure their GHG emissions tend to be larger companies (with greater exposure to various climate-related transition risks by virtue of their size and economic footprint) as observed in our own baseline analysis (see Table 5) and in the assessments of commenters, many of whom supported exemptions for SRCs and EGCs as they would be disproportionately impacted by the requirement.[2842] While these exemptions may limit the benefit of achieving greater consistency and comparability across registrants, exempting SRCs and EGCs from this disclosure requirement at this time is appropriate given the relatively larger burden GHG emissions reporting requirements could have on these firms[2843] and the differences in the existing levels of climate-related disclosure between larger companies and smaller companies.

Commenters raised concerns about the costs of providing GHG emissions on a disaggregated basis.[2844] However, many commenters also explained that disaggregated disclosures could be decision useful, as emissions from specific constituent gases could have

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[2842] *See, e.g.*, letters from U.S. SBA ("Small entities will need to allocate larger shares of their technological, financial, and staff resources to comply with the proposed rules. Representatives from the biotechnology, plastics, and equipment manufacturing industries have reported to Advocacy that small businesses in their industries have not traditionally tracked GHG emissions or other climate-related metrics. These businesses would either need to develop modeling software to track climate metrics in-house or hire third-party consultants to do so … Small private companies have also voiced that the costs of collecting and analyzing GHG emissions data could be prohibitive."); Soros Fund (suggesting that EGCs and SRCs should be allowed additional time to adjust to climate disclosure requirements, be afforded an additional safe harbor and be exempt from financial statement metrics disclosure); and SBCFAC Recommendation (recommending "scaling and delaying the compliance requirement for emerging growth companies, along with smaller reporting companies.").

[2843] Even for SRCs and EGCs that are currently calculating GHG emissions, there could be certain fixed costs associated with preparing this information for disclosure in Commission filings that would not scale with the size of the registrant and would therefore be more burdensome to these entities. We expect benefits to scale with the size of the firm.

[2844] *See, e.g.*, letters from ABA; ERM CVS; Sullivan Cromwell; and T. Rowe Price.

differential effects on a company's cash flows or business operations.[2845] For example, a registrant may be subject to methane fees by the EPA, in which case information about the registrant's methane emissions could factor into investors' decision making. To balance these views, some commenters suggested that the final rules should require constituent gases to be disclosed on a disaggregated basis only when individually material.[2846] We agree with those commenters and believe that this approach will provide investors with decision-useful information about GHG emissions without imposing undue compliance costs on registrants to produce disaggregated data in circumstances in which the disaggregation may not be particularly useful for investors.[2847]

The final rules also permit registrants to calculate and disclose GHG emissions according to the methodology that best matches their particular facts and circumstances. The benefit of this flexible approach is that registrants will have the opportunity to provide investors with information about their GHG emissions using the latest and most suitable methodology as measurement technologies and standards continue to develop. For example, while many companies calculate their GHG emissions pursuant to the GHG Protocol, others utilize different approaches, such as certain ISO standards.[2848] This flexibility, which may include registrants'

2845 *See, e.g.*, letters from PwC; and WRI.

2846 *See, e.g.,* letter from Deloitte & Touche ("Many emissions category calculation methods are estimate-based and rely on proxy data; the potential variance in actual can be significant and is largely unknown in many instances. Especially given these challenges, the Commission may consider whether the disaggregated data by each constituent greenhouse gas should only be required to be disclosed when individually material.").

2847 *Id.*

2848 *See* letters from Futurepast (referencing ISO 14064-1, Specification with Guidance at the Organization Level for Quantification and Reporting of Greenhouse Gas Statements; and ISO 14067, Carbon Footprint of Products—Requirements and Guidelines for Quantification); and ISO Comm. GHG; *see also, e.g.*, letters from Alphabet *et al.*; As You Sow; Beller *et al.*; CalSTRS; CFA; Dell; Deloitte & Touche; Engine No. 1; ERM CVS; KPMG; Morningstar; Soc. Corp. Gov.; and WRI.

ability to round as appropriate, will serve to limit costs.[2849] Conversely, it could also make comparisons less straightforward, which may attenuate some of the expected benefits of the final rules. However, there are several reasons to believe that this reduction in comparability will not significantly undermine the utility of the required disclosures. First, the required disclosures will expand upon and enhance the quality of the existing set of GHG emissions disclosures that investors already find useful despite the variation in methodologies that produce existing emission disclosures.[2850] Second, the contextual disclosures (e.g., operational boundaries) will enable investors to better understand the quantitative disclosures and make adjustments to facilitate comparisons with other registrants that are otherwise not possible under the baseline. Third, to the extent that industry-specific approaches to disclosing emissions continue to develop and evolve, the final rules will permit registrants within those industries to adopt those approaches, which will help investors to compare peer companies within an industry. Finally, as we discuss in the next subsection, obtaining assurance over GHG emissions disclosure provides investors with an additional degree of reliability regarding not only the figures that are disclosed, but also the key assumptions, methodologies, and data sources the registrant used to arrive at those figures.

These disclosures complement the other required disclosures about the organizational boundaries (used to calculate emissions versus those used in their financial statements) as well as carbon offsets and RECs, which offer important context for facilitating comparisons between companies as discussed above. In fact, by not requiring organizational boundaries to necessarily conform to those used in the company's consolidated financial statements, the final rules permit

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[2849] *See* letter from AGs of TX *et al.*

[2850] *See, e.g.*, letters from Vanguard; Fidelity; BlackRock; CALSTRS; and CALPERS for investors who derive utility from existing emissions disclosures.

the development of a standardized framework (*e.g.*, control approach) for measuring emissions across registrants. Commenters supported this approach as it would allow registrants to continue to measure emissions using their current approach and procedures.[2851] That is, by not imposing a prescriptive methodology for GHG emissions disclosures, the final rules provide space for the continued development of a shared reporting framework for issuers to disclose information that ultimately may enhance the degree of comparability of registrant-level GHG emissions data, to the benefit of investors, registrants and the market (relative to the baseline).[2852]

Finally, as discussed in section II.I above, we are following the suggestions of many commenters and allowing registrants more time to report emissions given the inherent challenges with reporting sooner that commenters highlighted.[2853] By delaying the requirement to disclose GHG emissions until later in the year, the final rules will provide additional time to prepare the information for filing (more consistent with current voluntary reporting practices),[2854] which should improve its accuracy and reduce costs for registrants but may result in delayed disclosure in some instances. The delay in annual reporting may also allow registrants to leverage disclosures they may have already prepared for other reporting regimes. Nonetheless, even with the extended filing deadline for registrants, investors will still benefit from receiving this information in a more timely and predictable manner than they currently do.[2855]

2851 *See, e.g.*, letters from API; ACORE; AHLA; and Chevron.

2852 *See, e.g.*, letters from Alphabet *et al.*; and Alliance-Bernstein.

2853 *See*, *e.g.*, letters from Morningstar; and American Banker.

2854 *See*, *e.g.*, *supra* note 2570 (stating "many companies still take more than 12 months after their fiscal year to disclose their sustainability data").

2855 *See, e.g.*, letter from Morningstar ("Currently, a lack of clear disclosure standards for the timing of 'sustainability reports,' which is the primary source for emissions data, greatly hinders investor knowledge. For example, some registrants released 2021 reports—detailing 2020 data—as late as November 2021.").